

ANYmal-D Kamino DVI Tuning

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****Outcome: stopped with no safe finalist.**** No Stage-2 candidate satisfied every per-seed learning guardrail. The ANYmal-D preset was not modified; final and canonical runs were skipped. No winner or winner speedup is reported.
Stage-2 metrics use two-sided 95% Student-t confidence intervals with n=2. Wave 1/2 rankings are single-seed observations without confidence intervals. Runtime excludes iterations 1--10; reward, TensorBoard Metrics/success_rate, and episode length use the final 20 iterations.

Early-stop decision

- Candidate: None
- Environments: 4096
- Resolved configuration: null
- Canonical comparison: null
- Terminal status: stopped_no_safe_finalist
- Stop reason: No Stage-2 candidate satisfied every per-seed learning guardrail.

Stage funnel

Stage	Attempted runs	Valid runs	Terminal-rejected runs	Learning-rejected candidates	Promoted candidates
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baseline	3	3	0	0	1
Wave 1	18	16	2	0	6
Wave 2	6	6	0	0	6
halve	16	16	0	8	0
final	0	0	0	0	0
canonical	0	0	0	0	0

Stage 2 also selected 2 candidate(s) that originated in Wave 1; these are not counted as Wave 2 promotions.

Stage-2 metrics (95% CIs)

Candidate	Runtime [s]	Reward	Success	Episode length
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block_iterations_4	0.721 ± 0.003 (95% CI, n=2)	1.438 ± 2.045 (95% CI, n=2)	0.299 ± 0.515 (95% CI, n=2)	215.532 ± 40.870 (95% CI, n=2)
block_iterations_8	0.790 ± 0.016 (95% CI, n=2)	1.931 ± 2.311 (95% CI, n=2)	0.329 ± 0.269 (95% CI, n=2)	272.632 ± 162.550 (95% CI, n=2)
combined_top_02	0.695 ± 0.035 (95% CI, n=2)	1.015 ± 0.240 (95% CI, n=2)	0.261 ± 0.457 (95% CI, n=2)	183.855 ± 201.358 (95% CI, n=2)
combined_top_03	0.533 ± 0.054 (95% CI, n=2)	-0.212 ± 1.105 (95% CI, n=2)	0.050 ± 0.239 (95% CI, n=2)	6.531 ± 19.583 (95% CI, n=2)
combined_top_04	0.524 ± 0.025 (95% CI, n=2)	-0.100 ± 0.037 (95% CI, n=2)	0.085 ± 0.087 (95% CI, n=2)	5.353 ± 3.501 (95% CI, n=2)
combined_top_05	0.521 ± 0.011 (95% CI, n=2)	-0.102 ± 0.099 (95% CI, n=2)	0.083 ± 0.098 (95% CI, n=2)	5.538 ± 4.320 (95% CI, n=2)
combined_top_06	0.521 ± 0.025 (95% CI, n=2)	-0.216 ± 1.281 (95% CI, n=2)	0.043 ± 0.260 (95% CI, n=2)	5.747 ± 11.683 (95% CI, n=2)
combined_top_07	0.517 ± 0.025 (95% CI, n=2)	-0.204 ± 1.170 (95% CI, n=2)	0.047 ± 0.299 (95% CI, n=2)	5.713 ± 11.293 (95% CI, n=2)

Contextual legacy runtimes

Backend	Runtime [s]	Comparison status
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mjwarp	0.370925	contextual only
physx	0.479481	contextual only

No apples-to-apples winner speedup is available because no candidate passed Stage 2.

Stability, failures, and rejections

- block_iterations_4: seed 42: reward below 80% of baseline
- block_iterations_8: seed 42: episode length ratio outside [0.8, 1.2]
- combined_top_02: seed 42: reward below 80% of baseline
- combined_top_03: seed 42: reward below 80% of baseline
- combined_top_04: seed 42: reward below 80% of baseline

Audit details (continued)

- combined_top_05: seed 42: reward below 80% of baseline
- combined_top_06: seed 42: reward below 80% of baseline
- combined_top_07: seed 42: reward below 80% of baseline
- contact_block_preconditioner_true: preflight:numerical
- dynamics_preconditioning_true: preflight:crash
- preflight_contact_block_preconditioner_true_seed42_env4096_iter5_attempt0: preflight:numerical
- preflight_dynamics_preconditioning_true_seed42_env4096_iter5_attempt0: preflight:crash

Methodology and provenance

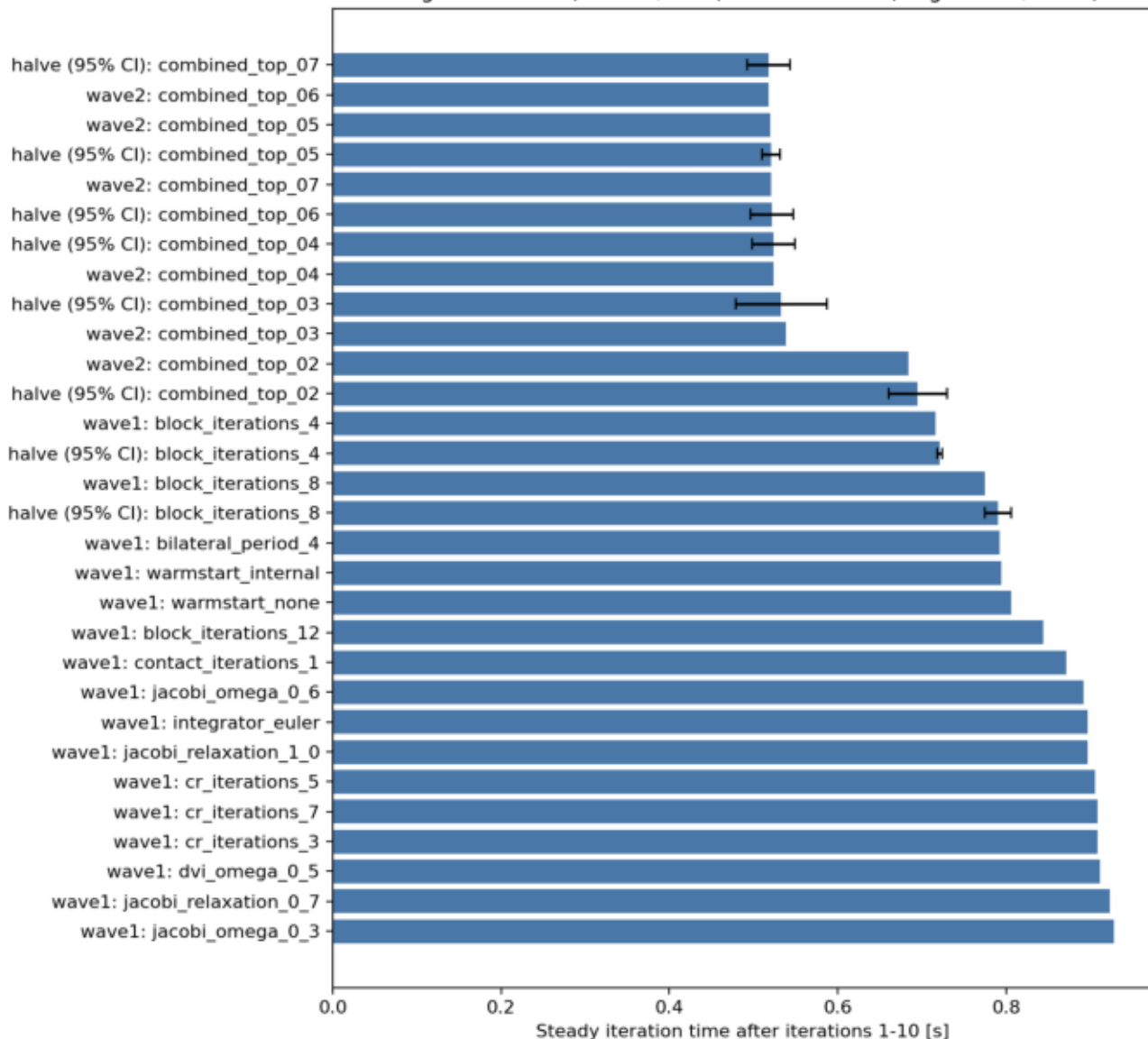
- Environment and coverage: 4096 environments; baseline seeds 42--44 at 300 iterations; Wave 1/2 seed 42 at 40; halve seeds 42--43 at 100; final and canonical skipped (zero attempts)
- Stage 2 baseline: first 100 aligned iterations of clean 300-iteration baseline; final-20 is iterations 81--100; source manifest/event/config hashes are retained in finalists.json
- Broad bundle dirty flags: 41; run IDs: baseline_baseline_seed42_env4096_iter300_attempt0, baseline_baseline_seed43_env4096_iter300_attempt0, baseline_baseline_seed44_env4096_iter300_attempt0, halve_block_iterations_4_seed42_env4096_iter100_attempt0, halve_block_iterations_4_seed43_env4096_iter100_attempt0, halve_block_iterations_8_seed42_env4096_iter100_attempt0, halve_block_iterations_8_seed43_env4096_iter100_attempt0, halve_combined_top_02_seed42_env4096_iter100_attempt0, halve_combined_top_02_seed43_env4096_iter100_attempt0, halve_combined_top_03_seed42_env4096_iter100_attempt0, halve_combined_top_03_seed43_env4096_iter100_attempt0, halve_combined_top_04_seed42_env4096_iter100_attempt0, halve_combined_top_04_seed43_env4096_iter100_attempt0, halve_combined_top_05_seed42_env4096_iter100_attempt0, halve_combined_top_05_seed43_env4096_iter100_attempt0, halve_combined_top_06_seed42_env4096_iter100_attempt0, halve_combined_top_06_seed43_env4096_iter100_attempt0, halve_combined_top_07_seed42_env4096_iter100_attempt0, halve_combined_top_07_seed43_env4096_iter100_attempt0, wave1_bilateral_period_4_seed42_env4096_iter40_attempt0, wave1_block_iterations_12_seed42_env4096_iter40_attempt0, wave1_block_iterations_4_seed42_env4096_iter40_attempt0, wave1_block_iterations_8_seed42_env4096_iter40_attempt0, wave1_contact_iterations_1_seed42_env4096_iter40_attempt0, wave1_cr_iterations_3_seed42_env4096_iter40_attempt0, wave1_cr_iterations_5_seed42_env4096_iter40_attempt0, wave1_cr_iterations_7_seed42_env4096_iter40_attempt0, wave1_dvi_omega_0_5_seed42_env4096_iter40_attempt0, wave1_integrator_euler_seed42_env4096_iter40_attempt0, wave1_jacobi_omega_0_3_seed42_env4096_iter40_attempt0, wave1_jacobi_omega_0_6_seed42_env4096_iter40_attempt0, wave1_jacobi_relaxation_0_7_seed42_env4096_iter40_attempt0, wave1_jacobi_relaxation_1_0_seed42_env4096_iter40_attempt0, wave1_warmstart_internal_seed42_env4096_iter40_attempt0, wave1_warmstart_none_seed42_env4096_iter40_attempt0, wave2_combined_top_02_seed42_env4096_iter40_attempt0, wave2_combined_top_03_seed42_env4096_iter40_attempt0, wave2_combined_top_04_seed42_env4096_iter40_attempt0, wave2_combined_top_05_seed42_env4096_iter40_attempt0, wave2_combined_top_06_seed42_env4096_iter40_attempt0, wave2_combined_top_07_seed42_env4096_iter40_attempt0
- Bundle dirty advisory: Broad/advisory bundle flag from plain git status --porcelain, which includes untracked paths; the runner separately enforced tracked-only cleanliness before launch. A true flag does not prove that only untracked paths differed.
- Derived preflight rejections: 2; sources: preflight_contact_block_preconditioner_true_seed42_env4096_iter5_attempt0 (preflight:numerical), preflight_dynamics_preconditioning_true_seed42_env4096_iter5_attempt0 (preflight:crash)
- Failed exact seed-42 screening preflights are projected in memory as rejected Wave 1/2 records only when measured evidence is absent; measured evidence always wins.
- Legacy comparison limitation: MJWarp/PhysX values come from the existing five-variant campaign whose manifests lack current exact source-HEAD and retained TensorBoard-event hashes; values are contextual only and no winner speedup is claimed.

Figures

![Runtime ranking](runtime.png)

![Learning guardrails](learning.png)

Stage-2 runtime (95% CI, n=2) and Wave 1/2 (single seed; no CI)



Stage 2 learning guardrails (including derived clean baseline)

